

Hadil Ftini

Embedded Systems Engineer | Robotics | AI

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PROFILE

Electrical Engineering Student and Embedded Systems Graduate. Passionate about embedded technologies, artificial intelligence, and software development. I have a particular interest in designing intelligent solutions that integrate hardware and software, especially in IoT, computer vision, and real-time systems.

PROFESSIONAL EXPERIENCE

EMBEDDED SYSTEMS INTERN

January – February 2026

Tunitech Engineering Services

- Studied and developed the architecture of an industrial automated marking machine.
- Designed a vision-based quality inspection system using computer vision and OpenCV.
- Integrated a Siemens S7-1200 PLC for machine control and monitoring.
- Implemented communication between the PLC, Raspberry Pi, and vision system using Modbus TCP/IP and serial communication.
- Conducted tests and validated the hardware and software architecture.

EMBEDDED SYSTEMS DEVELOPER

February – May 2024

IrWise – Startup

- Developed a LoRa communication monitoring application on Raspberry Pi.
- Designed a graphical user interface using PyQt to display RSSI and SNR data in real time.
- Implemented serial communication and data acquisition.

EDUCATION

ENGINEERING DEGREE IN ELECTRICAL ENGINEERING – MICROELECTRONICS 2025 – Present

Polytech Monastir

MASTER'S DEGREE IN EMBEDDED SYSTEMS

2024 – 2026

Faculty of Sciences of Monastir

BACHELOR'S DEGREE IN EMBEDDED SYSTEMS

2021 – 2024

Faculty of Sciences of Monastir

HIGH SCHOOL DIPLOMA IN MATHEMATICS

2021

Lycée 9 Avril 1938

TECHNICAL PROJECTS

Automated Marking Machine with Computer Vision System

Siemens S7-1200 – Raspberry Pi – Python – OpenCV – Modbus TCP/IP

- Studied the architecture of an automated marking machine integrating a vision-based inspection system.
- Developed a computer vision application for recognizing and verifying printed characters.
- Implemented communication between the PLC and the computer vision application.

Autonomous Mobile Robot Development Using ROS

ROS – Python – Gazebo – RViz – SLAM – AMCL – LiDAR

- Developed an autonomous mobile robot using ROS, with simulation in Gazebo and visualization in RViz.
- Implemented environment mapping using SLAM and a LiDAR sensor.
- Integrated robot localization using AMCL and autonomous navigation using move_base.
- Implemented obstacle detection and automatic path replanning for obstacle avoidance.
- Developed and tested a ROS architecture integrating nodes, topics, services, and TF transformations.
- Validated autonomous navigation in a simulated environment using Gazebo and RViz.

Intelligent Voice Assistant

Python – YOLOv5 – OpenCV – Kivy – ESP32

- Developed an intelligent assistant integrating speech recognition and object detection.
- Integrated an ESP32 for hardware control.

- Developed a graphical user interface using Kivy.
- [Project on GitHub](#)

LoRa Communication Monitoring Application

Python – Raspberry Pi – RFM96/98 – PyQt

- Developed an application for monitoring LoRa communications.
- Implemented real-time visualization of RSSI and SNR parameters.
- [Project on GitHub](#)

OCR-Based Text Recognition System

Python – OpenCV – PyTesseract – Tkinter

- Developed an application for text recognition from images using OCR.
- Designed a graphical user interface using Tkinter.

Smart Display System on STM32

STM32F4 – C – ILI9341 – SX1280

- Developed firmware for an ILI9341 TFT display using the STM32F4 microcontroller.
- Implemented LoRa communication using the SX1280 module.

TECHNICAL SKILLS

Programming Languages: C, C++, Python, Embedded C

Embedded Systems and Microcontrollers: STM32, ESP32/ESP8266, Raspberry Pi, Arduino

Robotics and Middleware: ROS 1, SLAM, AMCL, Gazebo, RViz, Autonomous Navigation

Computer Vision and Artificial Intelligence: OpenCV, YOLO, TensorFlow, PyTorch, Tesseract OCR

Communication Protocols: UART, SPI, I2C, Modbus TCP/IP, MQTT, LoRa

GUI Development: PyQt, Tkinter, Kivy

Version Control: Git

Operating Systems: Linux

Computer-Aided Design (CAD): SolidWorks, CATIA